

NLP • HEALTHCARE • HUMAN-CENTRED AI

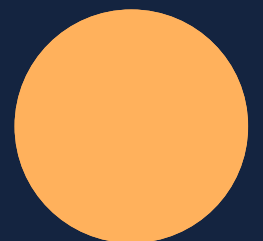
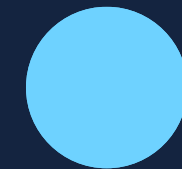
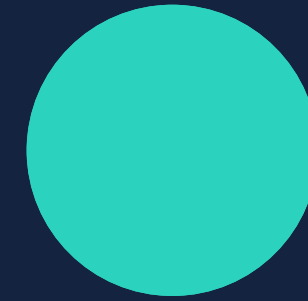
MEDPLAIN

Making complex medical language
clearer for everyday readers



A rule-based Natural Language Processing project

PROJECT PRESENTATION | Python • Regex • Gradio



The idea in one slide

Translate difficult clinical wording into simpler language, then make the result easier to inspect.

THE PROBLEM

Medical notes can contain abbreviations, technical terms and instructions that are hard for non-specialists to understand.

THE APPROACH

A transparent, rule-based NLP pipeline applies dictionaries, regular expressions and sentence-pattern replacements.

THE EXPERIENCE

A Gradio interface presents simplified text, detected topics, difficult terms and basic text statistics.

Core principle: improve readability without pretending to replace a clinician.

Why medical language needs a plain-language layer

The project focuses on comprehension—not diagnosis or treatment decisions.

COMPLEX INPUT

**“Discontinue immediately...
contraindicated in patients with
hypertension.”**

PLAIN-LANGUAGE GOAL

**“Stop right away...
should not be used by people with high blood
pressure.”**

Potential value: clearer instructions, faster scanning, and more accessible health information.

What the system is designed to do

The notebook contains both a text-simplification pipeline and an interactive analysis layer.

-
- Expand common medication-frequency abbreviations such as qd, bid, tid, qid and qhs.
 - Replace selected medical terms with familiar everyday wording.
 - Rewrite common instruction patterns into more direct sentences.
 - Detect broad topics such as dosing, monitoring, warnings and follow-up.
 - Highlight difficult terms and report basic text statistics in the interface.

Dataset and preparation

The notebook expects a CSV file named medplain_10k.csv with complex, plain and category fields.

complex

Original medical wording that the system attempts to simplify.

plain

Reference plain-language text used for comparison where available.

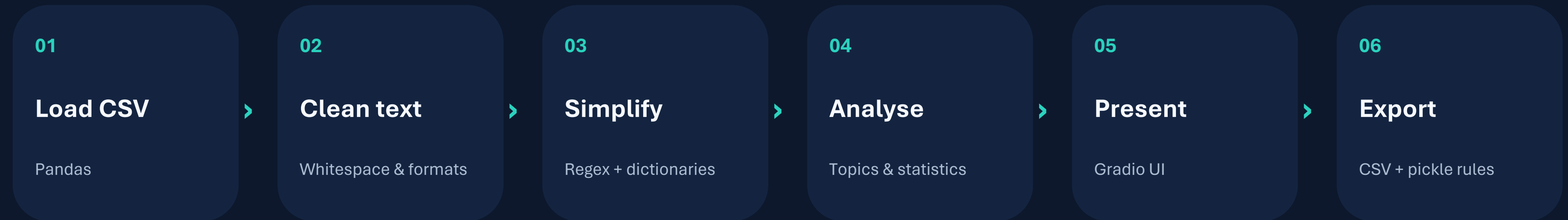
category

Topic labels used for category-count exploration and topic evaluation.

Preparation steps in the notebook: load CSV → inspect shape and missing values → strip surrounding whitespace → examine category counts.

End-to-end workflow

A deterministic pipeline makes transformations visible and easy to debug.



The design favours understandable, hand-authored rules over a trained neural language model.

How simplification works

The first pipeline applies transformations in a deliberate order.

- Normalize dotted abbreviations (for example, b.i.d. → bid).
- Rewrite monitoring instructions into reader-friendly phrases.
- Apply sentence-level patterns for warnings and care instructions.
- Replace known medical terms with simpler equivalents.
- Expand frequency abbreviations and normalize spacing/capitalization.

```
# Illustrative rules from the notebook
frequency = {
    r'\\bq\\.?.?d\\.?.?\\b': 'once a day',
    r'\\bb\\.?.?i\\.?.?d\\.?.?\\b': 'twice a day',
    r'\\bq\\.?.?h\\.?.?s\\.?.?\\b': 'at bedtime'
}
```

Longest phrases are replaced first to reduce partial-match problems.

Dictionaries turn jargon into familiar wording

Examples are drawn from the medical vocabulary in the notebook.

hypertension

→ high blood pressure

dyspnea

→ difficulty breathing

edema

→ swelling

sutures

→ stitches

oral

→ by mouth

PRN

→ as needed

The notebook includes a broad dictionary, abbreviation expansion, and reusable sentence patterns.

Beyond rewriting: inspect the message

The second layer extracts simple signals that help users review the text.

TOPIC DETECTION

Rule-based keyword checks for dosing, stopping medicine, follow-up, monitoring, warnings, diagnosis and diet/care.

DIFFICULT TERMS

Finds terms in the configured dictionary and displays their plain-language meanings.

TEXT STATISTICS

Counts words, sentences, characters and detected hard words; splits text into sentences.

These are heuristic signals: topic labels and readability indicators should be treated as aids, not authoritative classifications.

A Gradio interface for interactive review

The notebook builds a web-style interface around the text analysis function.

MEDPLAIN • Medical text analyser

Paste medical text here...

Analyse text

Output: simplified wording • topics • hard terms • statistics

- Highlights difficult words in original text.
- Shows simplified output.
- Displays detected topic chips.
- Lists term meanings and sentence-level details.

What the notebook measures

The notebook calculates evaluation values at runtime; no fixed scores are assumed in this presentation.

EXACT-MATCH COMPARISON

Compares the generated simplified text with the dataset's plain column after lowercasing. This is strict: semantically equivalent wording may still count as a mismatch.

TOPIC DETECTION ACCURACY

Compares predicted topic labels with category labels from the dataset. This depends on the quality and consistency of keyword rules and annotations.

Important: report the actual output from a complete run, including dataset version and evaluation setup, before claiming performance.

Strengths—and what must be handled carefully

A credible project presentation should explain where the current approach can fail.

STRENGTHS

- Transparent, explainable rules
- Lightweight; no model training required
- Easy to extend dictionaries
- Interactive analysis workflow
- Exports reusable rules and results

LIMITATIONS

- Rules may miss unseen wording
- Replacements can alter context or grammar
- Abbreviations can be ambiguous
- Exact match is not full clinical validation
- Safety-critical instructions need expert review

Medical text simplification must preserve dosage, timing, contraindications and warning meaning.

How the project could be strengthened

A practical next-step plan moves from reliability to richer evaluation.

01 • Validate

Review outputs with clinicians or qualified medical-language reviewers.

02 • Test robustly

Create test cases for spelling, punctuation, abbreviations and unseen terms.

03 • Measure meaning

Add readability metrics and semantic similarity alongside exact match.

04 • Protect safety

Flag uncertain transformations and preserve numbers, doses and urgent warnings.

05 • Compare approaches

Benchmark rules against a validated NLP or language-model baseline.

TAKEAWAYS

Clearer language. More informed review.

- MedPlain combines medical-term replacement, abbreviation expansion and sentence patterns.
- The analysis layer adds topic hints, difficult-word detection and simple text statistics.
- The current system is a useful prototype; clinical safety and effectiveness require systematic validation.